

The proposed upper pseudomode curve is not optimal for the linear imaginary biharmonic potential

Candidate counterexample to the optimality question in arXiv:2201.03305

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Status: candidate full negative answer to the universal/full-line optimality claim, likely valid. For

$$H = \frac{d^4}{dx^4} + ix \quad \text{on } L^2(\mathbb{R}),$$

there are pseudomodes at every $\lambda = \alpha + i\beta$ with $\alpha \rightarrow +\infty$, uniformly in $\beta \in \mathbb{R}$. Their residual is at most

$$C_\varepsilon \alpha^{1/16} \exp \left[- \left(\frac{8}{5} - \varepsilon \right) \alpha^{5/4} \right].$$

In particular, $\alpha = \beta^3$ gives pseudomodes far beyond the source's upper curve $\alpha \lesssim \beta^{8/3-o(1)}$ for $V(x) = ix$. Thus that curve is not an optimal boundary for the pseudospectrum of the original full-line operator.

1 The source question

Nguyen Duc studies the maximal full-line operator

$$H_V = \frac{d^4}{dx^4} + V(x)$$

and constructs pseudomodes near a positive-end turning point x_β , defined by $\text{Im } V(x_\beta) = \beta$. Theorem 2.2 assumes, among other conditions,

$$[\beta \tau(x_\beta)]^{4/5} \lesssim |\alpha| \lesssim [\beta \tau(x_\beta)^{-1}]^{(4+\varepsilon_1)/(3+\varepsilon_1)}. \quad (1)$$

Immediately afterward, the paper states that when $\text{Re } V = 0$ it expects both bound curves in (1) to be optimal and calls their optimality an open problem [1, p. 7, Theorem 2.2 and the following paragraph]. Here is the complete source context.

extended on $\mathbb{R}^+$, i.e. $|J_\beta| = 4\Delta_\beta = 4\eta x_\beta \rightarrow +\infty$ as $\beta \rightarrow +\infty$. This is one of the strengths of this direct construction which makes it going beyond the semi-classical construction (see [20, Remark 2.3]). Now we can state our second theorem:

Theorem 2.2. *Let Assumption II hold for some $N \in \mathbb{N}_0$. Assume that there exists a (β -dependent) α such that the following holds as $\beta \rightarrow +\infty$, for all $x \in J_\beta$,*

$$\alpha - \operatorname{Re} V(x) \approx |\alpha|, \quad (2.16)$$

$$[\beta \tau(x_\beta)]^{\frac{4}{5}} \lesssim |\alpha| \lesssim [\beta \tau(x_\beta)^{-1}]^{\frac{4+\varepsilon_1}{3+\varepsilon_1}}. \quad (2.17)$$

Then, there exist $c > 0$, $\beta_0 > 0$ and a family $(\Psi_{\lambda,N}) \subset \operatorname{Dom}(H_V)$ such that for all $\beta \geq \beta_0$, we have

$$\frac{\|(H_V - \lambda)\Psi_{\lambda,N}\|_{L^2(\mathbb{R}^+)}}{\|\Psi_{\lambda,N}\|_{L^2(\mathbb{R}^+)}} \lesssim \kappa(\beta) + \sigma^{(N)}(\beta),$$

in which

$$\begin{aligned} \bullet \quad \kappa(\beta) &:= \exp \left(-c \frac{\operatorname{Im} V^{(1)}(x_\beta) \tau(x_\beta)^{-2}}{|\alpha|^{\frac{3}{4}} + \beta^{\frac{3}{4}}} \right), \\ \bullet \quad \sigma^{(0)}(\beta) &:= \sum_{k=0}^2 \sum_{j=1}^{k+1} \frac{\tau(x_\beta)^{k+1} (|\operatorname{Re} V(x_\beta)|^j + \beta^j)}{|\alpha|^{\frac{k-3}{4}+j}}, \\ \bullet \quad \sigma^{(N)}(\beta) &:= \sum_{k=0}^{3N-1} \sum_{j=1}^{k+N+1} \frac{\tau(x_\beta)^{k+N+1} (|\operatorname{Re} V(x_\beta)|^j + \beta^j)}{|\alpha|^{\frac{k+N-3}{4}+j}}, \quad N \geq 1. \end{aligned}$$

The same result for Schrödinger operator given in [21, Cond. (5.5)] is known to be optimal when considering the case $\operatorname{Re} V = 0$, we believe that in this case our bound curves for α in (2.17) is also optimal. The study of optimality of our estimates on these bounds constitutes an interesting open problem.

2.1.3. *Semi-classical setting.* Let us consider the semi-classical biharmonic operator on $\mathbb{R}$

$$H_\beta = h^4 \frac{d^4}{dx^4} + W(x)$$

We show that the proposed upper curve is not a necessary boundary, even for the admissible purely imaginary polynomial $V(x) = ix$, provided the question is read for the original full-line operator.

2 Proof intuition

The linear imaginary potential has an exact symmetry that the local WKB analysis does not use. Fourier transform sends multiplication by ix to $-\partial_\xi$, so $H - (\alpha + i\beta)$ becomes

$$-\partial_\xi + \xi^4 - \alpha - i\beta.$$

The phase $e^{-i\beta\xi}$ removes β exactly. For $\alpha = a^4$, the remaining first-order equation has the exact homogeneous solution

$$e^{\Phi_a(\xi)}, \quad \Phi_a(\xi) = \frac{\xi^5}{5} - a^4 \xi.$$

Its exponent has a local maximum $\frac{4}{5}a^5$ at $\xi = -a$ and a local minimum $-\frac{4}{5}a^5$ at $\xi = a$. We retain the maximum and cut the solution off just after the minimum. The norm is controlled by the

maximum, while the cutoff error is controlled by the minimum, producing the barrier $\frac{8}{5}a^5 = \frac{8}{5}\alpha^{5/4}$. This construction is uniform in the imaginary spectral coordinate.

3 Uniform full-line pseudomodes

Let H be the maximal operator associated with $u \mapsto u^{(4)} + i xu$ on $L^2(\mathbb{R})$. Every Schwartz function belongs to $\text{Dom}(H)$.

Theorem 1. *For every $\varepsilon > 0$ there are constants $C_\varepsilon > 0$ and $A_\varepsilon > 0$ such that, for every $\alpha \geq A_\varepsilon$ and every $\beta \in \mathbb{R}$, there is a nonzero $u_{\alpha,\beta} \in \mathcal{S}(\mathbb{R})$ with*

$$\frac{\|(H - (\alpha + i\beta))u_{\alpha,\beta}\|_2}{\|u_{\alpha,\beta}\|_2} \leq C_\varepsilon \alpha^{1/16} \exp\left[-\left(\frac{8}{5} - \varepsilon\right)\alpha^{5/4}\right]. \quad (2)$$

Consequently, if $\alpha_n \rightarrow \infty$, then $\alpha_n + i\beta_n$ has pseudomodes for every real sequence (β_n) .

Proof. Use the unitary Fourier transform with $\widehat{xf} = i\partial_\xi \widehat{f}$. Then

$$\mathcal{F}(H - (\alpha + i\beta))\mathcal{F}^{-1} = -\partial_\xi + \xi^4 - \alpha - i\beta. \quad (3)$$

Put $a = \alpha^{1/4}$ and

$$\Phi_a(\xi) = \frac{\xi^5}{5} - a^4 \xi = a^5 F(\xi/a), \quad F(t) = \frac{t^5}{5} - t. \quad (4)$$

Thus $(e^{\Phi_a})' = (\xi^4 - \alpha)e^{\Phi_a}$.

Fix $\varepsilon > 0$. Since $F(1) = -4/5$, choose $\delta > 0$ so small that

$$F(1 + \delta) \leq -\frac{4}{5} + \varepsilon. \quad (5)$$

Choose $\chi \in C_c^\infty((-3, 1 + \delta))$ with $\chi = 1$ on $[-2, 1]$, and set

$$v_{a,\beta}(\xi) = e^{-i\beta\xi} \chi(\xi/a) e^{\Phi_a(\xi)}. \quad (6)$$

This is a nonzero compactly supported smooth function, so $u_{\alpha,\beta} = \mathcal{F}^{-1}v_{a,\beta}$ is Schwartz. Substitution into (3) gives the exact cancellation

$$(-\partial_\xi + \xi^4 - \alpha - i\beta)v_{a,\beta} = -a^{-1}e^{-i\beta\xi}\chi'(\xi/a)e^{\Phi_a(\xi)}. \quad (7)$$

We now compare the two sides. The phase has

$$F(-1) = \frac{4}{5}, \quad F'(-1) = 0, \quad F''(-1) = -4.$$

On the interval $|\xi + a| \leq a^{-3/2}$, Taylor's theorem therefore gives $\Phi_a(\xi) \geq \frac{4}{5}a^5 - C$ for all large a . This interval lies where $\chi(\xi/a) = 1$, hence

$$\|v_{a,\beta}\|_2 \geq ca^{-3/4}e^{\frac{4}{5}a^5}. \quad (8)$$

The constants do not depend on β because the extra phase has modulus one.

The support of χ' is contained in $(-3, -2) \cup (1, 1 + \delta)$. The function F is increasing on each of these intervals, and

$$F(-2) = -\frac{22}{5}, \quad F(1 + \delta) \leq -\frac{4}{5} + \varepsilon.$$

It follows from (7), after accounting for a support of length $O(a)$, that

$$\|(-\partial_\xi + \xi^4 - \alpha - i\beta)v_{a,\beta}\|_2 \leq C_\varepsilon a^{-1/2} e^{(-\frac{4}{5} + \varepsilon)a^5}. \quad (9)$$

Dividing (9) by (8), using Plancherel, and recalling $a = \alpha^{1/4}$ yields

$$\frac{\|(H - (\alpha + i\beta))u_{\alpha,\beta}\|_2}{\|u_{\alpha,\beta}\|_2} \leq C_\varepsilon a^{1/4} e^{-(\frac{8}{5} - \varepsilon)a^5} = C_\varepsilon \alpha^{1/16} e^{-(\frac{8}{5} - \varepsilon)\alpha^{5/4}},$$

which is (2). The final assertion is immediate. $\square$

4 Failure of the proposed upper curve

For $V(x) = ix$, the source's hypotheses hold with

$$x_\beta = \beta, \quad \tau(x) = (1 + x^2)^{-1/2}, \quad \tau(x_\beta) \asymp \beta^{-1}.$$

Thus the upper curve in (1) has order

$$[\beta \tau(x_\beta)^{-1}]^{(4+\varepsilon_1)/(3+\varepsilon_1)} \asymp \beta^{2(4+\varepsilon_1)/(3+\varepsilon_1)}, \quad (10)$$

whose exponent is strictly less than $8/3$. Take $\alpha = \beta^3$. Then (10) is violated by a power of β , whereas Theorem 1 gives

$$\frac{\|(H - (\beta^3 + i\beta))u_{\beta^3,\beta}\|_2}{\|u_{\beta^3,\beta}\|_2} \leq C_\varepsilon \beta^{3/16} \exp\left[-\left(\frac{8}{5} - \varepsilon\right)\beta^{15/4}\right] \rightarrow 0.$$

Therefore the proposed upper curve is not an optimal boundary for the full-line pseudospectrum, even in the purely imaginary linear case.

5 Checks, scope, and novelty

The argument is exact up to the elementary cutoff estimates: the Fourier identity, removal of β , and residual formula (7) contain no asymptotic expansion. The constructed Fourier functions are in C_c^∞ , so their inverse transforms lie in the maximal operator domain. The exponent $8/5$ is the height difference between the local maximum and minimum of the phase and can be approached arbitrarily closely.

This result answers the universal/full-line optimality belief negatively. It does *not* decide whether the same upper curve is optimal for a genuine half-line realization with a boundary condition at 0, or within the narrower class of pseudomodes required to be supported entirely in $\mathbb{R}^+$. Indeed, the source's positive-end WKB construction imposes precisely such localization, while the full-line symmetry is what removes β here. The lower curve in (1) is also not addressed.

Novelty check. Bounded searches through 11 August 2026 covered the exact source title and optimality wording, “biharmonic” or “fourth-order complex Airy” with “pseudomode” or “resolvent”, and the model $d^4/dx^4 + ix$. No later paper recording this Fourier-space counterexample was found. Searches did find the well-studied second-order complex Airy operator and later work on complex magnetic Laplacians, but neither answers this fourth-order curve-optimality question. Novelty confidence is moderate because the search was bounded.

Human-review recommendation. Check the Fourier sign in (3), the exact phase cancellation in (7), the $a^{-3/4}$ norm scale near $\xi = -a$, and the interpretation of the source’s optimality sentence as a claim about the full-line pseudospectrum rather than only positive-half-line-supported WKB pseudomodes.

References

- [1] T. Nguyen Duc, *Pseudomodes for biharmonic operators with complex potentials*, Asymptotic Analysis 132 (2023), no. 3–4, 451–493; arXiv:2201.03305, <https://arxiv.org/abs/2201.03305>.